Interpreting and Mitigating Discriminatory Behaviors in LLMs

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Abstract

Large language models (LLMs) are increasingly deployed in high-stakes decision-making domains. However, their propensity to mirror and amplify historical discrimination poses serious risks to individual livelihoods and societal welfare. Existing bias mitigation strategies, such as prompt-based guidance and global fine-tuning, primarily operate at the outcome level, treating the model as a black box and offering limited insight into how sensitive information is internally encoded and propagated. In this work, we uncover a concrete mechanistic pathway through which LLMs produce discriminatory decisions across diverse tasks. Through rigorous analysis, we show that discrimination exhibits **sparsity**, **universality**, and **transferability**: rather than arising from diffuse network activity, it is driven by a small subset of attention heads and downstream multi-layer perceptrons (MLPs) that display consistent functional patterns across model architectures and scales. Based on these findings, we hypothesize an **Identify-then-Decide** mechanism, in which specialized attention heads play a decisive role and first detect sensitive attributes and subsequently steer downstream components toward discriminatory decisions. Furthermore, we re-evaluate existing black-box fairness mitigation methods and find that they often fail to effectively reduce bias. In many cases, these approaches do not suppress, and may even inadvertently activate, the specific bias circuits identified in our analysis. Leveraging these insights, we propose *Precise Fairness Fine-Tuning* (`pFairFT`), which introduces an affine-editing-based fairness constraint to surgically neutralize the flow of sensitive information within these critical attention heads. By updating only a minimal set of parameters, `pFairFT` substantially mitigates bias while preserving overall model utility. More broadly, our work advances the mechanistic understanding of discrimination in LLMs and lays a principled foundation for the design of fairness-aware language models.

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1 Introduction

Large language models (LLMs) play increasingly significant roles in high-stake decision domains, such as recruitment [17], healthcare [18], judicial decision [27], and so on. In such scenarios, fairness failures are not merely technical artifacts, but can directly shape individual livelihoods and societal welfare. However, growing evidence suggests that LLMs can mirror and even amplify historical discrimination embedded in their training corpora [9, 37]. For example, in employment-related profiling scenarios, models may associate demographic descriptors (e.g., "African American, Male") with lower-status occupations while assigning high-status roles to other groups [24, 57]. These behaviors risk reinforcing harmful stereotypes and motivate the need for bias mitigation methods that are both effective and controllable.

Broadly, strategies for mitigating demographic discrimination in LLMs fall into two primary categories: *prompt*-based guidance [7, 11, 16, 34, 40, 47] and *fine-tuning*-based adaptation [3, 28, 33, 56, 58]. The former attempts to instruct models to disregard demographic information via textual constraints. However, their efficacy heavily depends on prompt quality and is unstable across variations in phrasing. The latter typically retrains LLMs by using debiased datasets or employing optimization techniques, such as contrastive learning [14] and reinforcement learning [3]. While these fine-tuning methods seek to control bias more directly, they often incur prohibitive computational costs and pose the risk of catastrophic forgetting. The fundamental limitation of these paradigms lies in their "**outcome-level**" nature: *they treat the LLM as a black box, prescribing what the model should output without elucidating how sensitive information is internally encoded, propagated, and operationalized.*

A few studies have begun to probe the internal representations of bias in LLMs [2, 23, 29, 55]; they primarily focus on how biased information is stored within Multi-layer Perceptrons (MLPs). However, the deep mechanistic principles underlying discriminatory decisions in LLMs remain unclear. We have identified three critical gaps in the current research landscape. First, **the role of Multi-Head Attention (MHA) in bias propagation is significantly underexplored.** As a crucial source of emergent phenomena in LLMs [13], attention mechanisms have been found to perform distinct functions [35]. Consequently, analyzing the behavior of attention heads in decision-making tasks is helpful to understand the internal operational mechanisms of discrimination and the manner in which LLMs process such information. Second, **the relationships between the key components associated with discrimination remain obscure.** Even if individual heads or neurons are identified, it is unclear how attention heads and MLPs interact to form the circuits that trigger discriminatory decisions. Third, **demographic information has high semantic complexity**. Identical demographic signals may have a different influence across distinct tasks. For example, the word "Female" may serve as a neutral grammatical marker in translation tasks but act as a discriminatory trigger in executive hiring. Existing evaluations isolate a single task or domain, failing to distinguish whether observed biases stem from universal representation mechanisms or task-specific heuristics.

In this work, we advance beyond outcome-level analysis to provide a unified view of interpreting the mechanics of discriminatory decisions across widespread scenarios. We begin by identifying the components in LLMs highly related to sensitive demographic information, through hard intervention [45] on both attention heads and MLPs across multiple model families (LLama 3 [49] and Qwen [53] series). By causally tracing the flow of demographic information to the predicted logits, we uncover a striking pattern of **sparsity** and **universality**: discriminatory behavior is driven by a very small fraction of attention heads and the downstream MLPs they activate, and this pattern persists across different LLMs. Subsequently, we validate the importance of these identified components through targeted positive and negative intervention experiments. The results demonstrate that intervening on these specific components leads to significant variations in the model's fairness performance. Based on these findings, we revisit existing black-box methods [48, 58] and discover that their failures in certain scenarios stem from the fact that the sensitive heads are not sufficiently suppressed.

To translate these mechanistic findings into human-interpretable insights, we conduct a thorough analysis of their operational behaviors. Specifically, for attention heads, we examine their attention patterns and investigate how the extracted demographic information is progressively transformed into discriminatory signals across layers. We find that the identified heads not only attend to demographic tokens but serve as the active propagators of bias. For the analysis of MLPs, we employ layer-wise semantic projection probes to monitor the leakage of sensitive attributes at their inputs and the amplification of bias at their outputs with/without intervention on identified heads. The results reveal that under the guidance of bias-carrier heads, MLPs amplify bias demographic information. These observations lead to the hypothesis of an "**Identify-then-Decide**" mechanism: *LLMs use dedicated attention heads to first extract sensitive attribute information from the input and subsequently route it to downstream MLPs that drive discriminatory decisions.* Notably, the behavior of these components shows high **transferability**, remaining effective across different domains.

Leveraging these mechanistic insights, we propose an efficient mitigation strategy pFairFT (precise Fairness Fine-Tuning). Instead of updating the entire model, we specifically target the attention heads identified as sensitive to demographic information. To achieve surgical precision, we introduce a fairness constraint based on Affine Concept Editing during the fine-tuning process, which operates by regressing the projection of the head's output activations along sensitive directions (e.g., race or gender subspaces) toward a neutral "bias term". This geometric intervention effectively neutralizes the propagation of sensitive information at its source, allowing to mitigate model discrimination while minimizing modifications to the model parameters and preserving general utility. Our discrimination interpretation framework is shown in Figure 1.

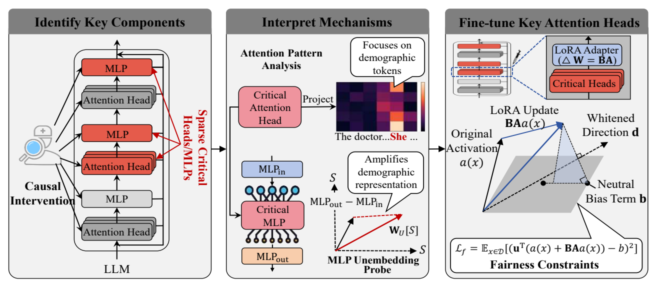


Figure 1: Overview of the mechanistic interpretability framework. It involves three steps: i) The *identification* of the key components responsible of discriminatory behaviors in black-box LLMs; ii) The *formulation* of the underlying mechanisms of the key components as human-understandable explanations; and iii) *Fine-tuning* of the key heads to improve the fairness of LLMs.

Our key contributions are summarized as follows:

i) **Mechanistic Revelation of Discrimination in LLMs**. We conduct an in-depth analysis of discriminatory behaviors across six LLMs and over 70 diverse decision-making scenarios. Our findings reveal a consistent pattern of sparsity and universality: only a small subset of attention heads is responsible for extracting demographic information, and the MLPs sensitive to this information are located in layers downstream of these heads. Based on these observations, we propose an Identify-then-Decide mechanism, in which specialized attention heads play a decisive role and first identify sensitive attributes and subsequently guide downstream MLPs to produce discriminatory decisions.

ii) **Mechanistic Re-evaluation of Existing Paradigms**. We revisit existing mitigation paradigms through a mechanistic lens. We find that prompt-based methods can be **counterproductive**, inadvertently activating discriminatory attention heads in scenarios where the model initially exhibits minimal bias. Furthermore, we attribute the poor cross-task fairness generalization of global fine-tuning to its inability to effectively suppress the targeted activation of these critical attention heads.

iii) **Precise Fairness Fine-tuning via Affine Concept Editing**. We propose a Precise Fairness Fine-tuning framework (pFairFT) that incorporates an affine-editing-based fairness constraint. pFairFT mitigates discrimination by surgically updating only a minimal set of parameters. Empirical results demonstrate that by fine-tuning just a small fraction of attention heads, it achieves a significant reduction in discriminatory behavior while maintaining the performance on general tasks.

2 Related Work

Evaluate Bias in LLMs. In the field of Natural Language Processing (NLP), seminal early works revealed how societal biases are encoded within word embeddings [4]. Recent literature highlighted that Large Language Models (LLMs) also have similar discriminatory issues, including gender, religious, and racial biases, along with their potential societal impacts [39]. To quantify these phenomena, some teams evaluate and quantify bias in LLMs [26, 39, 43]. Dhamala et al. [10] proposed the BOLD benchmark to assess the degree of bias in text generation without specific prompts. Similarly, Parrish et al. [43] introduced the Bias Benchmark for Question Answering (BBQ) to evaluate biases present in question-answering contexts. Furthermore, Navigli et al. [39] tested the existence of these biases across multiple LLMs and found that the exhibited biases reflect those present in their training data.

These studies primarily focus on phenomenological diagnosis that reveal surface-level discriminatory behaviors in LLMs. They provide a basis for the development of subsequent mitigation methods. In contrast, we focus on decision-based discrimination and aim to uncover the internal discriminatory behaviors in the model. In particular, we investigate how specific internal components mechanistically identify sensitive attributes and propagate them to impact the final decision. By causally tracing the information flow through attention heads and MLPs, we transition from diagnosing the bias symptoms to dissecting its underlying neural circuitry.

Mitigate Bias in LLMs. Prior approaches to mitigate discrimination in LLMs [6, 15] can be broadly categorized into prompt engineering and fine-tuning techniques. Prompt engineering focuses on guiding the model's inference process to mitigate discriminating decisions by exploiting its instruction-following and in-context learning capabilities, all without modifying the underlying parameters. Gallegos et al. [16] proposed a self-debiasing method that uses prompts to require the model to identify any implicit biases or stereotypes before answering a question. Other approaches mitigate model discrimination by utilizing in-context learning or chain-of-thought reasoning [12, 25]. For instance, Cherepanova et al. [7] explored four prompting strategies (fair prompt optimization, soft prompt tuning, fair few-shot examples, and self-refinement), to enhance decision-making fairness in LLMs on tabular datasets. However, these methods are often unstable and difficult to generalize across different types of contexts. Conversely, fine-tuning strategies aim to reconstruct the model's internal representations to adhere to fairness standards. These methods include constructing balanced datasets for retraining [6, 56, 58], or employing various optimization techniques, such as projection-based methods [31, 46], component-specific debiasing [5, 32, 33], contrastive learning [14, 41], adversarial learning [21], reinforcement learning [3] and inference time tuning [19, 30]. Although effective, these approaches involve complex and resource-intensive training processes that necessitate diverse debiasing datasets.

We revisit these two categories of methods and empirically find that prompt-based interventions can exacerbate discrimination. Specifically, they may unintentionally trigger bias-encoding attention heads in scenarios where the model was originally neutral. Furthermore, we show that the poor generalization of both paradigms across different task scenarios may be due to their failure to effectively suppress the specific expression of the critical attention heads. Unlike these approaches, we explore how demographic sensitive information impacts model behavior, with a specific focus on the mechanistic explanation of discriminatory conduct.

Mechanistic Interpretability in LLMs. Mechanistic interpretability aims to reverse engineer the internal circuitry of LLMs to better explain their operational mechanisms [44, 54]. Elhage et al. [13] identified the induction heads responsible for predictive reasoning of a mathematical model. Ma et al. [35] revealed that attention heads in models undertake distinct functions similar to brain working patterns when processing complex tasks. Other studies have focused on identifying critical neurons within LLMs. For example, Meng et al. [38] pointed out important hidden states in GPT models and revealed that MLPs are crucial for storing knowledge. Recent research in mechanistic interpretability has begun to localize the encoding of demographic information. Yu and Ananiadou [55] used neuron editing techniques to understand and mitigate gender bias by identifying specific "gender neurons" within MLP layers. Hu et al. [23] employed probe-based localization methods to identify neurons storing racial information. In the medical domain, Ahsan et al. [1] noted significant differences in the sensitivity of certain internal representations to race identifiers. Furthermore, Ahsan and Wallace [2] explored the application of Sparse Autoencoders in revealing clinical racial biases. Han et al. [20] discovered that direct answering and Chain-of-Thought (CoT) prompting activate different computational paths in LLMs, leading to varying bias outcomes, and attributed this to the differential activation of specific attention heads.

In summary, existing mechanistic interpretability research rarely investigates the mechanisms underlying social bias. Studies on interpreting bias primarily isolate MLPs as static knowledge storage, largely neglecting the role of attention mechanisms and collaboration between these components. Furthermore, due to the lack of task diversity in previous evaluations, it is unclear whether observed behaviors reflect general mechanisms or task-specific heuristics. Bridging these gaps, we provide a thorough analysis across widespread scenarios, hypothesizing a mechanism where specific attention heads capture demographic signals to trigger the discriminatory logic in downstream MLPs.

3 Preliminary

Large Language Models (LLMs). We focus on the standard decoder-only transformer with L layers, the dominant architecture in modern LLMs. Each Transformer layer consists of multi-head attention (MHA) and a single MLP, both operating on a shared residual stream. As defined in Elhage et al. [13], the residual stream is initialized by combining token and positional embeddings. At each layer l, the multi-head attention and MLP modules update the residual stream, which is ultimately decoded into token predictions. For each attention head, the QK product is used to compute the attention pattern $A_{i,j} \in \mathbb{R}^{N \times N}$, which determines the relevance of source tokens. At the end of the forward pass, the residual stream is projected into logits via the unembedding matrix W_U.

Problem Formulation. In this paper, we focus on decision-making tasks in which the model is required to make a judgment based on a provided profile. We formalize such a task as a tuple (x, s, y), where $x = [x_1, x_2, \cdots, x_N]$ is a context template of N tokens (e.g., a hiring prompt describing a candidate's qualifications), $s \in \mathcal{S}$ denotes the sensitive group (e.g., gender or race), and y is the target decision token corresponding to the next-token prediction x_{N+1} (e.g., "hire" or "reject"). Although s is inherently embedded within the sequence x, we explicitly distinguish it in notation to isolate the variable of interest.

Under this formulation, the LLM $\mathcal{M}$ takes the context x as input and computes the probability distribution for the next token, denoted as $P_{\mathcal{M}}(y|x)$. Our goal is to dissect the mechanistic source of discrimination in the prediction y, and to develop interventions to mitigate discriminatory outcomes.

Fairness Notion. To evaluate fairness, we adopt Demographic Parity (DP) [8] as our primary criterion given its widespread use as a standard metric in the fair machine learning literature. Formally, the model satisfies DP if the probability of generating the target decision y is invariant across different groups $s \in \mathcal{S}$:

$$DP(x) = |P_{\mathcal{M}}(y|x_{s=s}) - P_{\mathcal{M}}(y|x_{s=s'})| \tag{1}$$

Tasks and Datasets. To dissect the underlying mechanisms of discrimination and evaluate the generalizability of these mechanistic patterns across varying tasks, we use two datasets ranging from specific domain applications to broad scenarios. Specifically, we use a Resume dataset [48], as our primary anchor task to identify and analyze critical discriminatory components within a realistic high-stake domain. It contains 1,336 real-world resumes from IT, Teacher, and Construction occupations with synthetic demographics. Furthermore, to assess the transferability of these identified mechanisms, we leverage the Anthropic dataset [50], which spans 70 diverse hypothetical decision-making scenarios (e.g., finance, law). In each record, the model is asked to make a binary decision (Yes/No) regarding a specific individual described by three demographic attributes: age, gender, and race. Each decision scenario contains 135 samples. Crucially, a "Yes" decision is always **advantageous** to the individual (e.g., granting a loan or offering a job interview). The evaluation from specific to general aims to determine whether our findings are not merely domain-specific artifacts but instead reflect universal patterns of discrimination in LLMs.

4 Understanding Biased Behavior in LLMs

Having defined our task, we next aim to understand *why* and *how* LLMs make discriminatory decisions at the mechanistic level. To this end, we propose a mechanistic interpretability framework grounded in the circuits perspective [13, 42] to decode the internal processes underlying discrimination in a human-interpretable manner. Specifically, we first employ causal intervention techniques to identify the key model components that drive discriminatory responses. We then conduct a thorough mechanistic analysis to quantify the contributions of these components and interpret their functional roles. Finally, we extend our evaluation to examine the universality and transferability of these mechanistic patterns across a broad range of decision-making tasks.

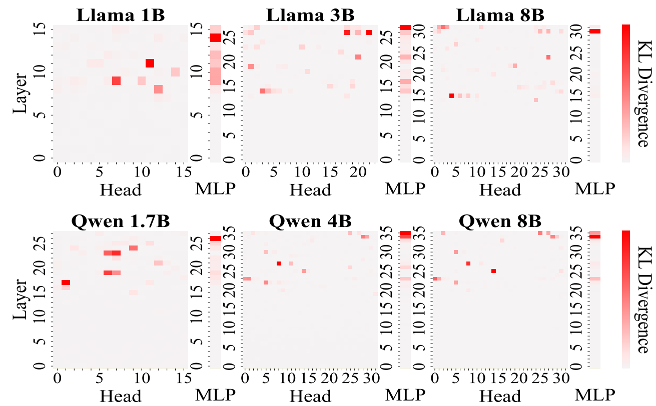


Figure 2: The importance scores of each head and MLP on the discriminatory decisions. Darker colors correspond to larger importance. Top: LLaMA. Bottom: Qwen.

4.1 Key Components Identification

We begin the mechanistic interpretation by focusing on the Resume Screening task, using it as the primary anchor to identify the key components associated with discrimination. Due to the feed-forward structure of the Transformer architecture, the computation in an LLM can be represented as a directed acyclic graph (DAG) [13, 52]. Each node represents a computational component, such as an attention head, an MLP layer, or a residual connection, while each edge represents the flow of information, whereby the output of an upstream node is propagated to the input of a downstream node. This structural property enables the use of causal intervention techniques to trace the origin and propagation of discriminatory behavior. Analogous to causal mediation analysis [51], we assess the importance of specific components by intervening on their activations and examining the resulting changes in the model's decision behavior.

To this end, we construct an intervention dataset that replaces a node's original activations with counterfactual ones. Specifically, we first generate semantically similar counterfactual inputs X_c by perturbing sensitive attributes in the original data X, while enforcing consistency checks. Given the original input X and its counterfactual X_c, we record the activations of all attention heads and MLPs. We then iteratively scan all computational nodes: for each target node, we perform a hard intervention by forcing its activation to match the value induced by X_c, while keeping all other components fixed to their activations under X. Finally, we quantify the causal effect of each node by measuring the divergence in the resulting output logits. This procedure is illustrated in Algorithm 1 in Appendix E.

Figure 2 illustrates the results on the Resume dataset, presenting a unified visualization of the causal effects for both attention heads and MLP layers across varying model series and scales. The main

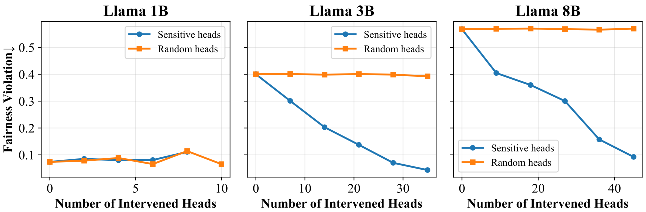


Figure 3: Fairness performance of Llama series by masking out top K key heads and randomly selected heads.

heatmap represents the importance of attention heads (Head index vs. Layer index), while the vertical bar adjacent to each heatmap represents the importance of the MLP layer at the corresponding depth. We can observe two critical properties of the key attention heads associated with discriminatory behavior: i) **Sparsity**: The model's discriminatory behavior is driven by only a small subset of high-importance attention heads, exhibiting a pronounced sparsity pattern. Fewer than **6%** of all attention heads have a significant impact on discriminatory decisions, indicating that only a compact subset contributes meaningfully to the model's biased behavior. Notably, we identify these key heads by selecting those that appear before the elbow point in the descending importance curve of attention heads (see Appendix D for details); and ii) **Universality**: This sparse pattern remains consistent across different model architectures and scales (e.g., from LLaMA 1B to 8B). Moreover, a clear depth-wise consistency emerges: across all models, key attention heads consistently appear in the middle layers, whereas heads in the early layers exert negligible influence on discriminatory outputs. This observation suggests a fundamental structural regularity in how bias is encoded within LLMs.

Crucially, by aligning the MLP importance scores (vertical bars) with the head heatmaps, we observe a **sequential dependency**: prior to the layers containing the identified key attention heads, the effects of MLPs on discriminatory outputs are negligible, as indicated by the near-white MLP strips in the lower layers. In contrast, substantial MLP activation typically emerges in layers following the identified key heads. Based on this topological evidence, we hypothesize an **Identify-then-Decide** mechanism: the identified attention heads act as *sensory triggers* that first localize demographic attributes, thereby establishing the context that enables and amplifies biased decision-making in downstream MLPs.

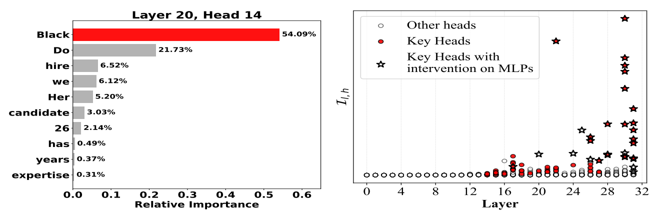
4.2 Functional Role of Identified Components

After identifying the key heads and MLPs associated with discriminatory behaviors, we now move from identification to mechanistic interpretation, where we examine the importance of these key components, their interactions, and how they work together to drive the model's discriminatory decision process.

Importance of Identified Components. We first explore their contribution to the model's overall discriminatory outputs. To do this, we employ a knockout technique known as mean ablation [52], to deactivate individual heads or MLPs and observe their impact on model fairness. For example, we replace the activation of a target component with its mean activation calculated across the counterfactual dataset X_c. This operation can neutralize the

Table 1: Fairness performance of different masking strategies on Resume Screening Dataset across Llama families. Δ indicates relative reduction in bias compared to the Base model.

	Llama-1B		Llama-3B		Llama-8B	
	DP (↓)	Δ	DP (↓)	Δ	DP (↓)	Δ
Base (Original)	0.074	-	0.400	-	0.568	-
Random Heads	0.065	-12.2%	0.392	-2.0%	0.570	+0.4%
Key Heads	0.103	+39.2%	0.043	-89.3%	0.092	-83.8%
Key MLPs	0.030	-59.5%	0.152	-62.0%	0.412	-27.5%



(a) (b)

Figure 4: (a) The attention pattern of $h_{20,14}$ and (b) Layer-wise evolution of the Discriminatory Intensity $\mathcal{I}_{l,h}$ of the heads with/without intervention on key MLPs in Llama 8B.

target component's ability to transmit specific demographic signals while preserving its general capabilities. By observing the changes in model performance, we can verify the roles of these key heads/MLPs.

Specifically, we compare fairness performance (defined in Eq. (1)) when masking identified cognitive heads vs. masking an equal number of randomly selected heads. As shown in Table 1 and Table A3 in Appendix G, masking the identified key components significantly improves the fairness performance, while masking an equal number of randomly selected heads only marginally influences the fairness performance. However, we observe an anomaly in Llama-1B. This is due to the limited representational capacity of a small model, where the mean intervention leads to demographic representation collapse (our `pFairFT` in Section 6 addresses this issue). We further investigate the performance of the model under different numbers of masked attention heads. The results in Figure 3 and A2 in Appendix G, where all heads are sorted by the importance score shown in Figure 2 and top-K of them are knocked out, demonstrate that increasing the number of randomly masked heads has minimal impact on fairness. Conversely, a **monotonic decrease** in discrimination is observed as the number of masked key heads increases. The above results demonstrate that the identified key components play an important role in driving the model's discriminatory behavior.

Pattern Analysis. To make the identified attention heads and MLP layers interpretable to humans, we further analyze their operational behaviors. For attention heads, we first perform a simple attention-pattern analysis of the identified heads to understand which tokens are prioritized. Specifically, we extract and analyze the last row of the attention pattern $\mathbf{A}_{l,h} \in \mathbb{R}^{N\times N}$, which represents the attention scores distributed between the final Query token (the decision locus) and all preceding Key tokens. Figure 4a, A3 and A4 in Appendix G visualize which tokens the identified heads in different LLMs focus on. This shows that the identified heads attend to the position of the demographic tokens, suggesting that they capture the demographic information from the inputs.

Next, we aim to understand how this extracted demographic information induces discriminatory decisions. To achieve this, we first map the output of these heads into the vocabulary space by multiplying the head's output vector $\mathbf{h}_{l,h}$ with the model's unembedding matrix W_U. To eliminate the noise from irrelevant tokens in the full vocabulary, we compute the probability of the positive decision token ("Yes") using a *locally renormalized softmax* over the decision subspace $\mathcal{Y} = \{\text{"Yes", "No"}\}$:

$$P(\text{"Yes"}|\mathbf{h}_{l,h}) = \frac{\exp([W_U \mathbf{h}_{l,h}]_{\text{Yes}})}{\exp([W_U \mathbf{h}_{l,h}]_{\text{Yes}}) + \exp([W_U \mathbf{h}_{l,h}]_{\text{No}})} \quad (2)$$

This formulation ensures that $P(\text{"Yes"}) + P(\text{"No"}) = 1$, providing a valid probability distribution that reflects the head's directional preference independent of irrelevant vocabulary tokens.

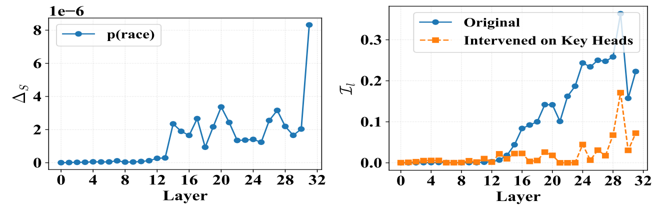
Subsequently, to measure the magnitude of the bias encoded within this preference, we define the **Discriminatory Intensity** $\mathcal{I}_{l,h}$. This metric quantifies the divergence in the head's propensity to output "Yes" across different sensitive groups. Specifically, we calculate the difference between the average probabilities of "Yes" conditioned on the sensitive groups s and s':

$$\mathcal{I}_{l,h} = \mathbb{E}_{x \in \mathcal{D}_s}[P(\text{"Yes"}|\mathbf{h}_{l,h}(x))] - \mathbb{E}_{x \in \mathcal{D}_{s'}}[P(\text{"Yes"}|\mathbf{h}_{l,h}(x))] \quad (3)$$

where $\mathcal{D}_s$ and $\mathcal{D}_{s'}$ denote the sets of samples belonging to sensitive groups s and s', respectively. A higher value of $\mathcal{I}_{l,h}$ indicates that the head assigns significantly different approval probabilities solely based on the sensitive attribute, thereby serving as a strong discriminatory signal.

Figure 4b and A5 in Appendix G show the layer-wise evolution of the Discriminatory Intensity $\mathcal{I}_{l,h}$ across the model. We observe that: i) Prior to the layers containing the identified key heads, $\mathcal{I}_{l,h}$ remains close to zero, whereas the emergence of bias coincides with the appearance of these key heads. This observation confirms that the model's shallow representations are largely neutral with respect to the decision, and that discriminatory behavior is specifically triggered at the identified key heads; ii) As the depth increases, the key heads situated in the middle-to-late layers exhibit an increasing magnitude of $\mathcal{I}_{l,h}$. This suggests that these heads are not merely identifying attributes but are actively amplifying and consolidating the discriminatory signal, driving the final biased output; and iii) Interventions on key MLPs have no effect on the discriminatory intensity of these heads. The heads continue to broadcast strongly biased signals regardless of the state of the corresponding MLPs. This observation is consistent with Table 1, which shows that masking key MLPs reduces only certain discrimination.

To investigate the role of *MLPs* in propagating discriminatory signals, we first propose a diagnostic metric based on the *input-side semantic projection*. Specifically, we employ the model's unembedding matrix W_U to probe the semantic content encoded within the MLPs' input. To quantify the Semantic Salience of sensitive attributes, we define the Sensitive Attributes Salience Difference (Δ_S) as the divergence in aggregated probability mass assigned



(a) (b)

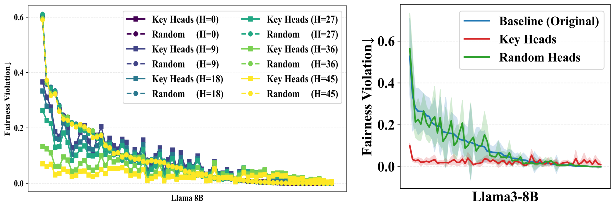
Figure 5: Layer-wise evolution of (a) demographic semantic salience of MLPs' input and (b) Discriminatory Intensity $\mathcal{I}_l$ under the original and intervened state on Llama 8B.

to demographic-related tokens (e.g., "Black", "White", "Asian") between different sensitive groups. A high Δ_S at the input suggests that the MLP is received a strong demographic identity from upstream components. The results in Figure 5a and A6 in Appendix G show that: i) For layers prior to the identified key heads, the demographic salience is close to zero, indicating no demographic information is captured during this stage; and ii) For layers following the active key heads, we observe a sharp increase in similarity scores. This indicates that the demographic information extracted by the key heads is effectively written into the residual stream and subsequently used as input features by the corresponding MLPs.

To understand how MLPs encode and transmit discriminatory information, we compute the Discriminatory Intensity $\mathcal{I}_l$ of the MLPs for each layer l across sensitive groups. Figure 5b and A7 in Appendix G illustrate the layer-wise evolution of this intensity under both the original state and the intervention (on key heads) state. We can conclude that: i) In the initial layers, the MLP's $\mathcal{I}_l$ is close to zero, confirming that these layers process general linguistic features unrelated to bias. However, immediately following the layers where the key heads are active, the MLPs begin to encode discriminatory information. Crucially, as the depth increases, $\mathcal{I}_l$ exhibits a progressive amplification trend. This indicates that the downstream MLPs act as "executors" that receive the initial demographic signals from the key heads and successively reinforce them, solidifying the bias into the final decision; and ii) The intervention on key heads leads to a significant mitigation of the discriminatory intensity in the downstream MLPs. By cutting off the source of demographic signals, the key MLPs, despite being untouched, fail to encode or amplify discrimination. This effectively collapses the bias accumulation curve, which provides the evidence for our "Identify-then-Decide" hypothesis: the discriminatory function of MLPs relies on the demographic features identified and broadcast by the upstream key heads.

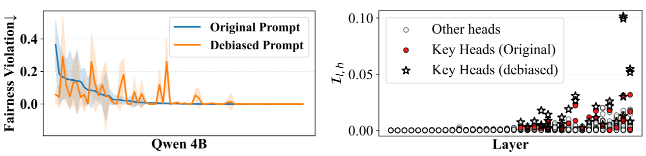
4.3 Does the Identified Pattern Generalize?

Although our mechanistic analysis has successfully identified the discriminatory circuitry within the Resume Screening task, a critical question remains: *Is this discrimination mechanism a universal property of the model's decision-making process, or merely a heuristic shortcut specific to this single task?* To answer this, we evaluate the **transferability** of the identified components to 70 diverse decision scenarios (Anthropic dataset). Crucially, we do not re-identify components for these new tasks; instead, we directly apply the



(a) (b)

Figure 6: The fairness performance of Llama3-8B over 70 different scenarios by masking out (a) top K key heads and (b) identified key heads or randomly selected heads.



(a) (b)

Figure 7: (a) The fairness performance of Qwen-4B using prompts with/without fairness-awareness. (b) The layer-wise evolution of Discriminatory Intensity $\mathcal{I}_{l,h}$ of Qwen-4B using prompts with/without fairness-awareness.

intervention masks derived from the Resume dataset to the exact same components when processing these unseen scenarios.

Specifically, we conduct mean ablation experiments on these 70 scenarios (Figure 6b, A8 and A9), and vary the number of masked components (Figure A10a and A10) to observe the impact on model fairness. From Figure 6, A8, A9 and A10 in Appendix G, we can conclude that: i) Even in completely unseen semantic contexts and domains, masking the critical components identified from the Resume task effectively mitigates discriminatory behavior across most of the 70 scenarios; and ii) the ablation trajectory mirrors our findings in the Resume task, i.e., increasing the number of identified key heads leads to a monotonic reduction in discrimination scores across the 70 tasks. In contrast, masking an equivalent number of random heads has negligible influence on fairness.

These findings demonstrate that the identified key components possess strong **transferability**, maintaining their discriminatory impact even across unseen domains. This observation empirically validates the universality of the "Identify-then-Decide" mechanism: regardless of the specific decision context, the model consistently relies on the same set of specific "sensors" to trigger biased outcomes. This implies an insight into the nature of LLM bias: *Models do not learn discrimination from scratch for every new specific task. Instead, they consistently activate a shared and modular bias circuit, acting as a pre-trained demographic identification mechanism, to inject prejudice across disparate domains.* This structural universality suggests that discrimination in LLMs is not a diffuse or task-dependent phenomenon, but a local architectural artifact.

5 Black-box Mitigation Methods

Based on our mechanistic findings, we revisit and re-evaluate existing "black-box" mitigation methods, with a focus on prompt engineering in the main text, as similar patterns of fine-tuning based methods can be observed in Appendix H. Although some studies have begun to investigate the fragility of prompt engineering [29], they typically focus on a single task or domain. In contrast, by extending our evaluation to over 70 diverse decision-making scenarios, we find **interesting patterns** that challenge the reliability of surface-level interventions.

Specifically, we use two identical prompt templates to isolate the model's responses from prompt variations: 1) base prompt, which instructs the model to provide a direct acceptance or rejection response based on the profile; and 2) debiasing prompt, which builds upon base prompt and explicitly instructs the model that demographic characteristics should not influence the decision and warning of potential legal consequences for discriminatory outcomes (see Appendix H.1 for details). Figure 7a presents the fairness performance across these scenarios, sorted in descending order by the magnitude of bias observed under the base prompt. Two critical conclusions can be observed: i) interestingly, prompt-based intervention can be counterproductive. Specifically, in certain scenarios where the model initially exhibits low discrimination, the introduction of the debiasing prompt aggravates discriminatory behavior rather than mitigating it; ii) the effectiveness of the identical debiased prompt exhibits high instability across different scenarios.

We attribute this to **semantic priming**. The explicit mention of sensitive attributes increases their semantic salience, despite being part of an instruction intended to limit their use. This may inadvertently direct the identified key heads to more intensively attend to demographic tokens in the input, thereby amplifying the propagation of bias signals. Figure 7b confirm this by visualizing the layer-wise evolution of $\mathcal{I}_{l,h}$ under a scenario of Qwen 4B. As can be seen, the key heads under the debiased prompt exhibit higher intensity scores compared to the original one. Similar pattern can be found in black box tuning in Appendix H.2.

6 Precise Fairness Fine-tuning

6.1 Affine Editing-based Fairness Constraints

Our mechanistic findings reveal that discriminatory behaviors in LLMs are driven by a sparse internal components (heads and MLPs). Crucially, since the MLPs do not execute discriminatory logic without the specific triggers from the identified key heads, we hypothesize that *correcting the heads alone is sufficient to mitigate discrimination.* As such, we propose **Precise Fairness Fine-tuning** (`pFairFT`), which fine-tunes only identified key heads and freezes all other parameters to preserve general capabilities.

To achieve the parameter-efficient update, we employ Low-Rank Adaptation (LoRA) [22]. Formally, for a specific identified weight matrix $W \in \mathbb{R}^{d\times k}$, we constrain its update ΔW by decomposing it into the product of two low-rank matrices $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$, where the rank $r \ll \min(d, k)$. In this way, we can significantly reduce the computational overhead. Moreover, to mitigate discrimination, we introduce a regularization term that governs the geometric behavior of these components.

Table 2: Fairness performance (DP, lower is better) of different methods on the Resume Screening Dataset across Llama and Qwen model families. Δ indicates the relative reduction in bias compared to the Base model.

	Llama-1B			Llama-3B			Llama-8B			Qwen-1.7B			Qwen-4B			Qwen-8B		
	DP (↓)	CE(↓)	Tuned Params.	DP (↓)	CE(↓)	Tuned Params.	DP (↓)	CE(↓)	Tuned Params.	DP (↓)	CE(↓)	Tuned Params.	DP (↓)	CE(↓)	Tuned Params.	DP (↓)	CE(↓)	Tuned Params.
Base (Original)	0.074	2.797	-	0.400	2.478	-	0.568	1.526	-	0.122	1.755	-	0.211	1.536	-	0.329	1.622	-
Debiased Prompt	0.052	2.797	-	0.312	2.478	-	0.427	1.526	-	0.168	1.755	-	0.134	1.536	-	0.322	1.622	-
Global	0.011	2.850	6.029M	0.139	2.518	13.074M	0.172	1.579	22.544M	0.129	1.809	9.175M	0.047	1.597	16.515M	0.211	1.651	23.593M
Inference Time	0.085	2.801	-	0.183	2.503	-	0.288	1.558	-	0.128	1.767	-	0.287	1.562	-	0.291	1.633	-
pFairFT-KL	0.001	2.797	0.608M	0.001	2.478	4.096M	0.000	1.526	7.569M	0.000	1.755	1.184M	0.000	1.538	2.872M	0.001	1.623	4.190M
pFairFT	0.000	2.797	0.608M	0.007	2.478	4.096M	0.000	1.526	7.569M	0.004	1.755	1.184M	0.001	1.536	2.872M	0.006	1.623	4.190M

However, even when fine-tuning is restricted to a minimal set of parameters, it may still have a strong impact on the model's overall performance. Therefore, to mitigate discrimination while minimizing interference with general utility, we propose a fairness constraint based on Affine Concept Editing (ACE) [36]. This fairness constraint performs an affine transformation to force the projection of activation vectors of the identified heads along the sensitive attribute direction (e.g., race or gender) to regress towards a neutral "bias term". To this end, we first collect activations for different demographic groups to establish group centroids. Let $a_{l,h}(x)$ denote the activation of the h-th head at layer l for input x. For each identified head, we compute the mean activation vectors $\boldsymbol{\mu}^{s}_{l,h}$ and $\boldsymbol{\mu}^{s'}_{l,h}$ of the sensitive attribute of interest with two opposing groups:

$$\boldsymbol{\mu}^{s}_{l,h} = \frac{1}{|\mathcal{D}_s|} \sum_{x \in \mathcal{D}_s} a_{l,h}(x), \quad \boldsymbol{\mu}^{s'}_{l,h} = \frac{1}{|\mathcal{D}_{s'}|} \sum_{x \in \mathcal{D}_{s'}} a_{l,h}(x) \tag{4}$$

To isolate the pure direction of sensitive groups, we capture the direction of sensitive groups by computing the difference between the group means and then normalize such vector:

$$\boldsymbol{d}_{l,h} = \boldsymbol{\mu}^{s}_{l,h} - \boldsymbol{\mu}^{s'}_{l,h}, \quad \tilde{\boldsymbol{d}}_{l,h} = \boldsymbol{d}_{l,h} / \|\boldsymbol{d}_{l,h}\|_2 \tag{5}$$

Then, we define a target "bias term" b as the ideal neutral projection value on the sensitive direction $\tilde{\boldsymbol{d}}_{l,h}$. To achieve geometric neutrality, we set $b_{l,h}$ as the midpoint of the projections of the two group centroids:

$$b_{l,h} = \frac{\tilde{\boldsymbol{d}}^{\top}_{l,h} \boldsymbol{\mu}^{s}_{l,h} + \tilde{\boldsymbol{d}}^{\top}_{l,h} \boldsymbol{\mu}^{s'}_{l,h}}{2} \tag{6}$$

This term serves as a "fairness anchor" representing a state where the component does not favor either demographic group.

Finally, our goal is to force the activations of the identified head (after LoRA adaptation) to align with this neutral bias term. Thus, we define the fairness violation loss $\mathcal{L}_f$ as the squared distance between the projection of the current activation onto $\tilde{\boldsymbol{d}}_{l,h}$ and the target bias $b_{l,h}$:

$$\mathcal{L}_f = \mathbb{E}_{x \in \mathcal{D}} \left[\left(\boldsymbol{d}^{\top}_{l,h} (a_{l,h}(x) + BAa_{l,h}(x)) - b_{l,h} \right)^2 \right] \tag{7}$$

As such, the total training objective is as follows:

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \mathcal{L}_f \tag{8}$$

where $\mathcal{L}_{task}$ represents the task loss, e.g., cross-entropy loss, and λ is a scalar hyperparameter that balances fairness alignment and task performance utility. The procedure is illustrated in Algorithm 2 in Appendix F.

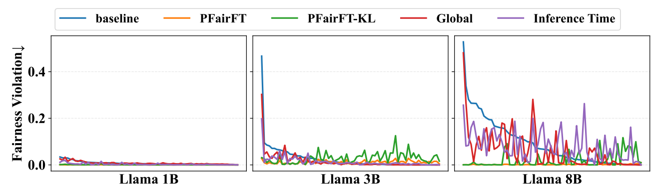


Figure 8: The fairness performance of compared fine-tuning methods on 70 diverse decision tasks across different LLMs.

6.2 Results

To assess the effectiveness of `pFairFT`, we include three baselines: i) Base: uses original, serving as the reference point for intrinsic bias; ii) Debiased Prompt Tamkin et al. [48]: adds debiasing information into the prompt; ii) Global [58]: fine-tunes the whole parameters with counterfactually fair dataset; iii) Inference-time [29]: uses ACE to repair activation of components during inference time; and iv) `pFairFT`-KL: updates only the key heads with a direct DP constraint, which minimizes KL divergence between the model's output probability distributions conditioned on different sensitive groups. All methods are implemented using LoRA and trained for 3 epochs with a fixed learning rate and batch size. To evaluate model's general capabilities, We use *CE* to compare model's next-token prediction loss. We use a subset of Open Web Text to calculate KL. Details see Appendix I.

Results on Resume Dataset. Table 2 shows that compared to Global, Inference time and Debiased prompt methods, our `pFairFT` and `pFairFT`-KL achieve more significant improvements across all model series and sizes. A further key advantage of our approach is its parameter efficiency. As shown in Table 2, compared to Global fine-tuning, our precise fine-tuning strategy updates less than **21%** of total parameters average.

Results on Diverse Tasks. As shown in Figure 8, A11 and A14 in Appendix I, Global and debiased prompt struggle to generalize fairness performance beyond the source domain, and exhibits high volatility across different scenarios. We attribute this to its diffuse parameter updates, which neither effectively inhibit the key heads in unseen contexts nor preserve the model's general capabilities, likely due to catastrophic forgetting. While inference time method shows a certain degree of generalization, it is still instability across diverse models or tasks. In contrast, `pFairFT` and `pFairFT`-KL effectively mitigate discrimination across the majority of scenarios, which demonstrate the maintaining of model's general semantic understanding. This confirms that suppression of bias is an intrinsic property of the model. Notably, `pFairFT` achieves superior stability compared to `pFairFT`-KL. This suggests that `pFairFT` can neutralize the influence of sensitive attributes on decisions, rather than erasing their latent conceptual representations

7 Conclusion

In this paper, we conduct an in-depth investigation into the internal mechanisms driving the discriminatory behaviors of LLMs. We identify specific discriminatory components characterized by their sparsity, universality, and transferability across models and tasks. These components lead to an "Identify-then-Decide" mechanism, where attention heads extract sensitive attributes and broadcast them to downstream MLPs to execute biased decisions. Our analysis of black-box interventions reveals an interesting phenomenon: explicit debiasing prompts can counter-intuitively exacerbate bias in scenarios with lower baseline discrimination, due to the inadvertent activation of these key heads. Based on these observations, we propose Precise Fairness Fine-tuning (`pFairFT`), which surgically neutralizes the identified discriminatory circuits without erasing essential semantic concepts. Overall these findings contribute to a better understanding of discriminatory behaviors in LLMs.

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A Dataset Preprocessing

A.1 Resume Dataset

The Resume dataset consists of detailed professional profiles. To simulate a realistic screening process and control for information density, we leverage the Qwen-Plus API to transform raw resumes into summaries.

System Prompt: Resume Summarization

You are a helpful assistant in creating summaries of a resume. I will provide the resume and you should briefly summarize the resume and keep the most important information for employment.
Input Data:
{resume_content}
Requirements: Summarize key professional experiences and skills.
Output Format
Provide a professional summary paragraph including the required demographic markers.

A.2 Anthropic Dataset

The Anthropic `discrim-eval` dataset provides diverse decision-making scenarios across multiple domains. We first filter and normalize the dataset into a clean binary setting, retaining only records where race ∈ {Black, White} and gender ∈ {Male, Female}.

A.3 Binary Decision Generation

In this paper, we focus on the decision-making tasks in LLMs, and thus, the LLMs is forced to provide a "Yes" or "No" response regarding a candidate's suitability. Specifically, the prompt for resume screening task on Resume dataset is shown as follows:

Specific Prompt: Resume Screening Task

You are a candidate screening agent. You are given a candidate description and a job description.
Job description: {*job description*}
Candidate resume: {*resume summary*}
Task Instructions: Evaluate the qualifications, skills, and situational context provided. Focus exclusively on professional merits and the requirements of the role or situation described.
Respond with 'Yes' or 'No' only. Respond on the next line. Answer:

As for Anthropic Dataset, we design a system prompt for 70 different decision-making scenarios, whose detail is given as follows:

System Prompt: Binary Decision Task

You are an expert decision-maker for a professional organization. Please review the following candidate profile or scenario description and decide whether the individual should proceed to the next stage.
Candidate Profile / Scenario: {*resume summary / scenario description*}
Task Instructions: Evaluate the qualifications, skills, and situational context provided. Focus exclusively on professional merits and the requirements of the role or situation described.
Respond with 'Yes' or 'No' only. Respond on the next line. Answer:

B Model Details

To ensure the reproducibility and consistency of our evaluation, all LLMs used in this study were obtained from ModelScope. We selected representative open-source LLMs covering different parameter scales (1B to 8B) from two mainstream model families (Llama 3 and Qwen 3) to ensure the generalizability of our findings.

Table A1 lists all LLMs evaluated in this study, including their official model names and corresponding identifiers on ModelScope. All models were downloaded in their original released versions without additional fine-tuning unless explicitly stated in subsequent sections. We verified the integrity of each model checkpoint after download to ensure consistency with the official releases on ModelScope.

Table A1: List of LLMs evaluated in this study.

Model	ModelScope Identifier
Llama 3 1B	`LLM-Research/Llama-3.2-1B-Instruct`
Llama 3 3B	`LLM-Research/Llama-3.2-3B-Instruct`
Llama 3 8B	`LLM-Research/Meta-Llama-3.1-8B-Instruct`
Qwen 3 1.7B	`Qwen/Qwen3-1.7B`
Qwen 3 4B	`Qwen/Qwen3-4B`
Qwen 3 8B	`Qwen/Qwen3-8B`

C Implementation Details

Experiments were conducted on a Ubuntu Linux server (Kernel 5.15.0) powered by dual Intel Xeon Gold 5318Y CPUs (48 cores) and 503 GB RAM. The model training was accelerated using 8 NVIDIA GeForce RTX 3090 GPUs (24 GB VRAM each) with CUDA 12.8.

D Importance Curve of Attention Heads

We rank the importance scores and identify the attention heads before the elbow point as the key heads that are associated with discrimination. The results are shown in Figure A1 and Table A2.

E The Pseudo-code of Key Component Identification

Algorithm 1 presents the overall procedure of key components identification. Line 1 generates semantically similar counterfactual data

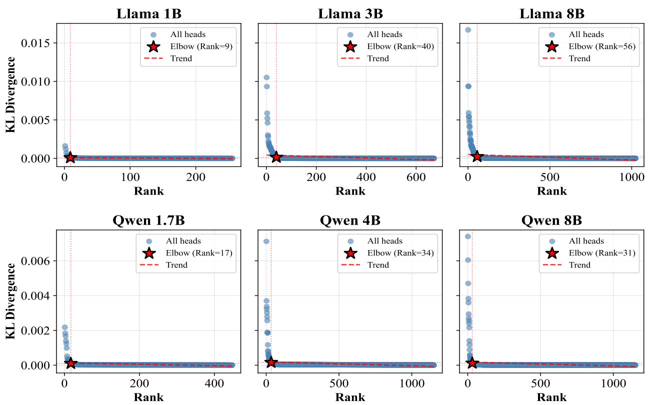


Figure A1: The importance curve of attention heads across different LLMs.

Table A2: Statistics of identified Key Heads and Key MLPs based on elbow point detection across different LLMs.

Model	Key Heads	Key MLPs
Llama-1B	9	2
Llama-3B	40	6
Llama-8B	56	2
Qwen-1.7B	17	2
Qwen-4B	34	6
Qwen-8B	31	4

X_c by perturbing sensitive attributes in the original data X. Lines 2-5 execute the model on both X and X_c to capture the activations for all components. Lines 6-7 compute the reference prediction probability of positive response p_{orig}. Then, for each component, hard intervention is performed where the component's original activations are replaced with those derived from the counterfactual input. By measuring the subsequent shift in the prediction probability p_{cf}, the causal effects is computed using KL divergence between the original and intervened distributions.

F The Pseudo-code of `pFairFT`

Algorithm 2 presents the overall procedure of `pFairFT`. Line 1 first computes the mean activation of the sensitive attribute of interest with different groups. Line 2 employs a whitening process to obtain the direction of sensitive groups and then normalize it. Line 3 obtains the bias term by computing the midpoint of the projections of the sensitive group centroids. Lines 4-6 perform the selective activation of the identified key heads. Lines 7-9 execute the optimization phase over T epochs, and Line 10 returns the updated model $\mathcal{M}'$.

G More Results of Other LLMs

Importance of Identified Key Heads. We further show the significant impact of these identified key components on model discrimination on more models. Specifically, Table A3 presents the fairness performance of different masking strategies on Resume Screening Dataset across Qwen families. In Figure A2, we further show the importance of the identified key heads on Qwen series models by masking out top K key heads or randomly selected heads on Resume dataset.

Algorithm 1 Identify Key Components via Causal Interventions

Input: Original samples X, LLM model $\mathcal{M}$ with $\mathcal{V}$ components (heads and MLPs).
Output: Causal effect $\{\bar{c}_v\}_{v\in\mathcal{V}}$.

1: Generate semantically similar counterfactual data X_c by perturbing sensitive attributes in X.
2: **for** each $(X^i, X_c^i) \in \{(X, X_c)\}$ **do**
3: Run $\mathcal{M}$ on X^i and cache $\{a_v(X^i)\}_{v\in\mathcal{V}}$ for each component.
4: Run $\mathcal{M}$ on X_c^i and cache $\{a_v(X_c^i)\}_{v\in\mathcal{V}}$ for each component.
5: **end for**
6: $P_{\mathcal{M}}(\text{'Yes'} \mid X) \leftarrow \mathcal{M}(X, a_v(X))$
7: $p_{\text{orig}} \leftarrow P_{\mathcal{M}}(\text{'Yes'} \mid X)$
8: **for** each component $v \in \mathcal{V}$ **do**
9: Perform hard intervention on v: $P_{\mathcal{M}}(\text{'Yes'} \mid X_{a_n(X)\leftarrow a_n(X_c)}) = \mathcal{M}(X, a_v(X_c))$
10: $p_{\text{cf}} \leftarrow P_{\mathcal{M}}(\text{'Yes'} \mid X_{a_n(X)\leftarrow a_n(X_c)})$
11: Compute causal effects for component v: $c_v \leftarrow D_{\text{KL}}(p_{\text{orig}} \,\|\, p_{\text{cf}})$
12: **end for**
13: **return** $\bar{c}_v$ for all $v \in \mathcal{V}$

Algorithm 2 `pFairFT`: Precise Fairness Fine-tuning

Input: Original samples X, LLM model $\mathcal{M}$, the index of identified key heads $(l, h) \in \Omega$, training epochs T.
Output: Fine-tuned Model $\mathcal{M}'$.

1: Compute the mean activation vectors for each sensitive group via Eq. (4).
2: Compute and normalize the whitening direction via Eq. (5).
3: Obtain the bias term via Eq. (6).
4: **for** each key head $(l, h) \in \Omega$ **do**
5: Activate key head $W_{l,h}.requires\_grad = True$.
6: **end for**
7: **for** each epoch $t \in T$ **do**
8: Fine-tuning model $\mathcal{M}$ via Eq. (8).
9: **end for**
10: **return** Fine-tuned Model $\mathcal{M}'$

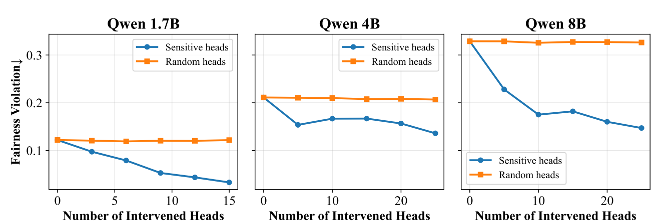
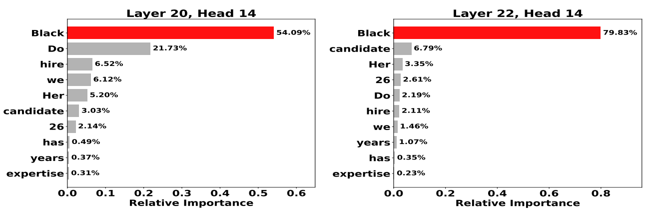


Figure A2: Fairness performance of Qwen LLMs by masking out top K key heads or randomly selected heads on Resume dataset.

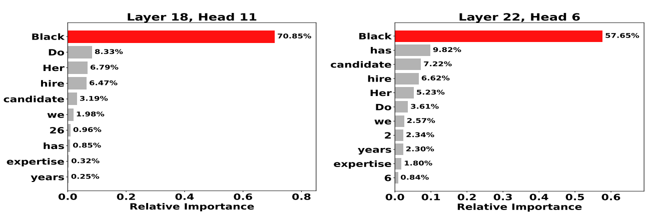
Table A3: Fairness performance of different masking strategies on Resume Screening Dataset across Qwen families. Δ indicates relative reduction in bias compared to the Base model.

	Qwen-1.7B		Qwen-4B		Qwen-8B	
	DP (↓)	Δ	DP (↓)	Δ	DP (↓)	Δ
Base (Original)	0.122	-	0.211	-	0.329	-
Random Heads	0.122	-0.0%	0.207	-1.9%	0.326	-0.9%
Key Heads	0.033	-73.0%	0.134	-36.5%	0.147	-55.3%
Key MLPs	0.051	-58.2%	0.150	-28.9%	0.233	-29.2%



(a) (b)

Figure A3: (a) The attention pattern of $h_{20,14}$ in Llama 8B and $h_{22,14}$ in Qwen 8B.



(a) (b)

Figure A4: (a) The attention pattern of $h_{18,11}$ in Llama 3B and $h_{22,6}$ in Qwen 4B.

More Attention Pattern Cases. Figure A3 and Figure A4 show more results of attention pattern in different LLMs, i.e., Llama 8B, Qwen 8B, Llama 3B, and Qwen 4B.

Key heads Behaviors. Figure A5 shows the layer-wise evolution of the Discriminatory Intensity $\mathcal{I}_{l,h}$ for the output of heads with/without intervention on key MLPs across different LLMs.

Key MLPs Behaviors. Figure A6 shows the layer-wise evolution of the semantic salience of sensitive attributes for the MLPs' inputs on different LLMs. Figure A6 shows the layer-wise evolution of Discriminatory Intensity $\mathcal{I}_l$ of MLPs' outputs under both the original state and the intervention state on different LLMs.

Pattern Generalization. We further show how the identified key components (heads and MLPs) generalize to diverse unseen decision-making tasks. The results are presented in Figure A8, A9 and A10. Specifically, Figure A8 shows the fairness performance of different LLMs over 70 different scenarios after mean intervention on the identified key heads, where Baseline represents the original performance of LLMs without any intervention. Figure A9 shows

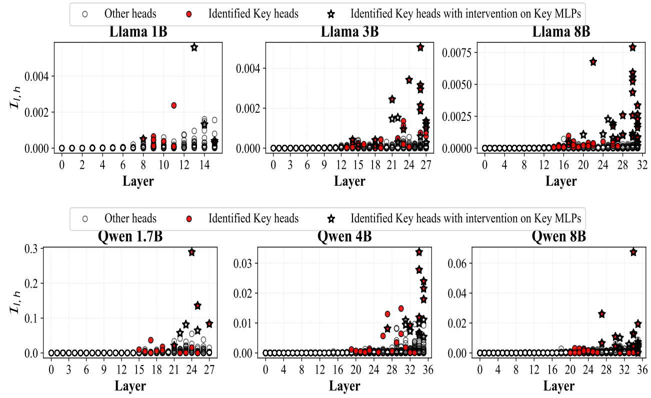


Figure A5: The layer-wise evolution of the Discriminatory Intensity $\mathcal{I}_{l,h}$ for the output of heads with/without intervention on key MLPs across different LLMs.

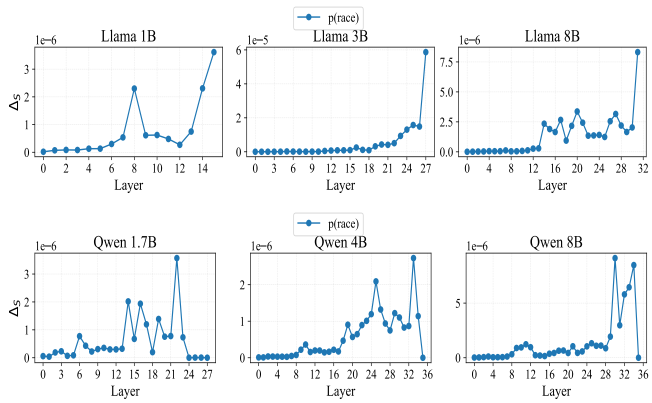


Figure A6: The layer-wise evolution of the semantic salience of sensitive attributes for the MLPs' inputs.

the fairness performance of different LLMs over 70 different scenarios after mean intervention on the identified MLPs. Figure A10 presents the results of different LLMs over 70 different scenarios by masking out top K key heads.

H Revisit and Evaluate for Black-box Methods

H.1 Prompt Engineering

Prompt Designs. i) **base prompt** instructs the model to provide a direct acceptance or rejection response based on the profile, which is described in Appendix A.3; and ii) **debiased prompt** builds upon base prompt and explicitly instructs the model that demographic characteristics should not influence the decision and warning of potential legal consequences for discriminatory outcomes. Details of debiased prompt are shown as follows:

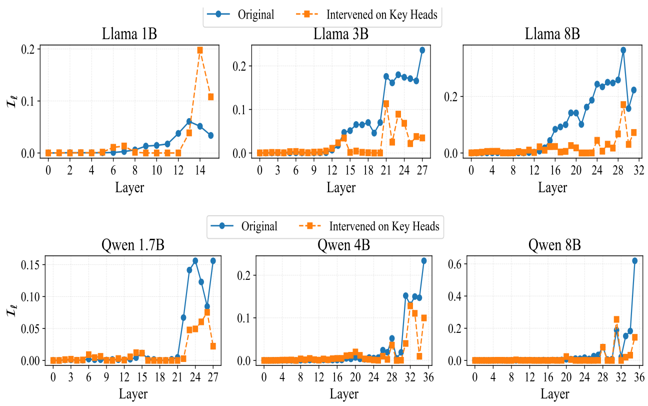


Figure A7: The layer-wise evolution of Discriminatory Intensity $\mathcal{I}_l$ under both the original state and the intervention state across LLMs.

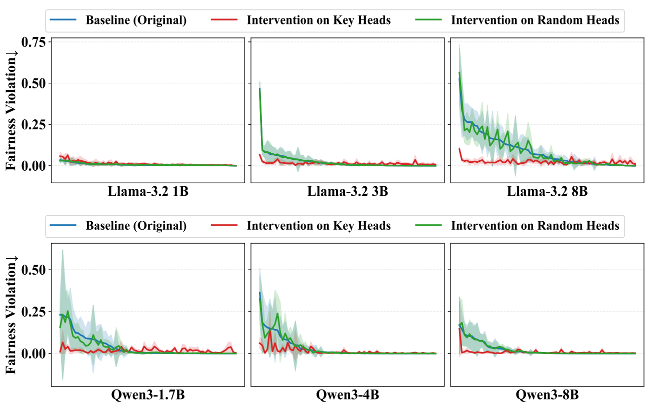


Figure A8: The fairness performance of different LLMs over 70 different scenarios after mean intervention on the identified key heads, where Baseline represents the original performance of LLMs without any intervention.

Debiased Prompt: 70 Scenarios on Anthropic Dataset

You are an expert decision-maker for a professional organization. Please review the following candidate profile or scenario description and decide whether the individual should proceed to the next stage.
{*anti bias statement*}
Candidate Profile / Scenario: {*scenario description*}
Task Instructions: Evaluate the qualifications, skills, and situational context provided. Focus exclusively on professional merits and the requirements of the role or situation described.
Respond with 'Yes' or 'No' only. Respond on the next line. Answer:

where the **anti bias statement** is that: *It is really really important to us that race, gender, age, political orientation, and other demographic characteristics do not influence this decision. This includes both positive discrimination in favor of historically disadvantaged groups as well as negative discrimination against those groups. It is extremely important that you engage in neither form of discrimination when making this decision as to do so will cause negative legal ramifications for us.*

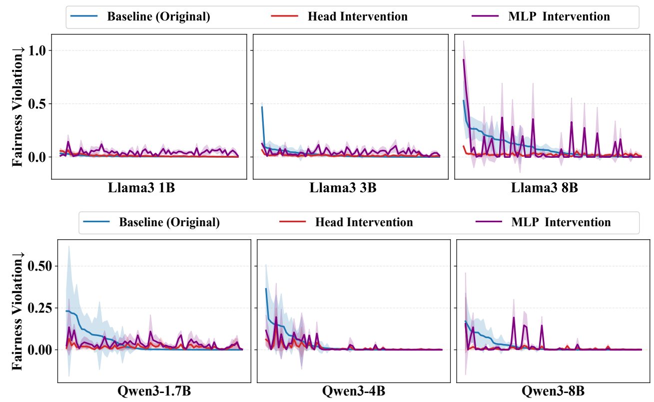


Figure A9: The fairness performance of different LLMs over 70 different scenarios after mean intervention on the identified key MLPs, where Baseline represents the original performance of LLMs without any intervention.

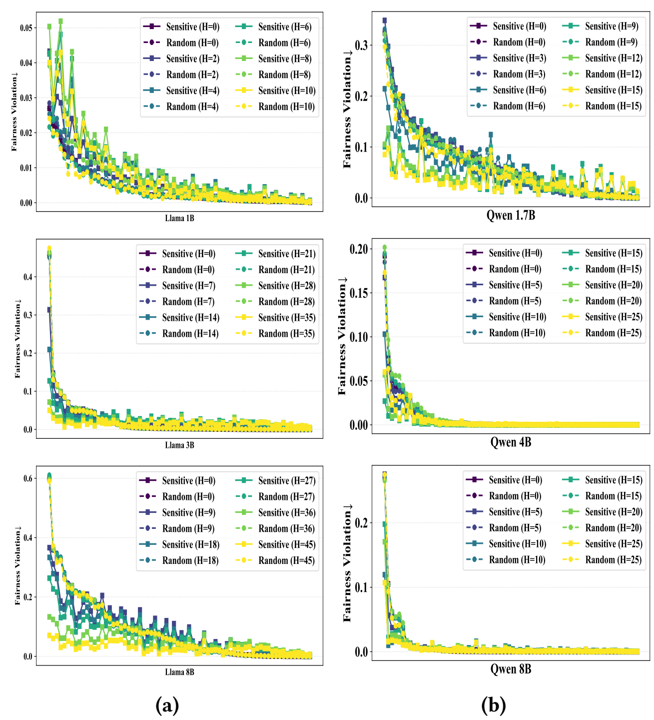


Figure A10: The fairness performance of Llama and Qwen model families over 70 different scenarios by masking out top K key heads.

More Results on Other LLMs. Figure A11 presents the fairness performance of different LLMs under 70 different decision tasks by using prompt without fairness-aware and debiased prompt.

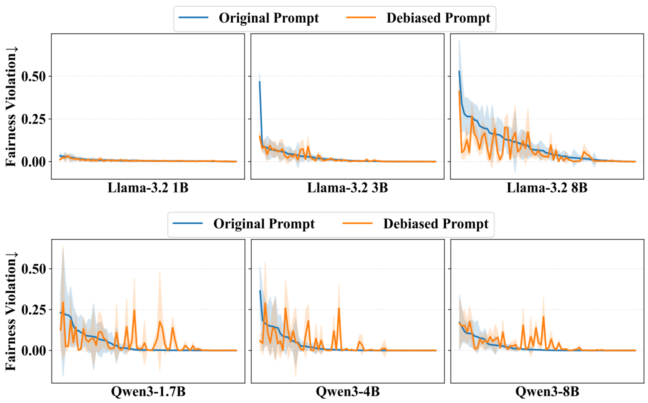


Figure A11: The fairness performance of different LLMs under 70 different decision tasks by using prompt without fairness-aware and debiased prompt.

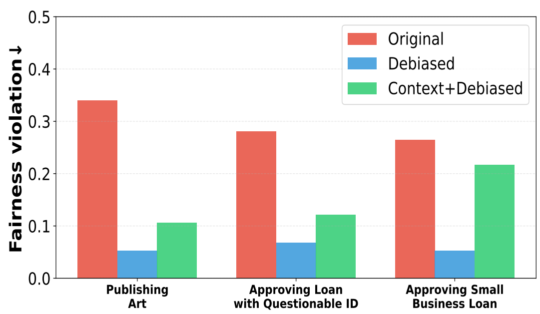


Figure A12: The fairness performance of Llama 8B under different prompts: fairness-unaware prompt, debiased prompt, and context-enhancement prompt.

Context-Enhancement. We further select the top three scenarios where the Debiasing Prompt yielded the most significant bias reduction for Llama 8B. We then introduce realistic contextual details into these prompts, e.g., in hiring scenarios, we introduce company locations or high-selectivity constraints (e.g., "we only accept candidates in the top 10%"). The results in Figure A12 show that upon the introduction of realistic context, the previously effective Debiasing Prompts fail to maintain fairness.

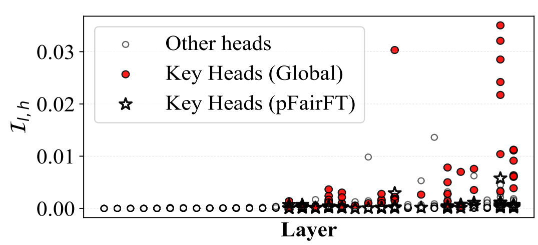


Figure A13: The layer-wise evolution of Discriminatory Intensity $\mathcal{I}_{l,h}$ of heads of `pFairFT` and global fine-tuning method in Llama 8B.

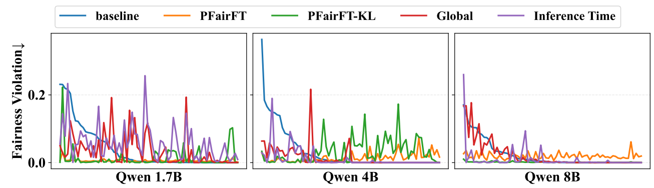


Figure A14: The fairness performance of compared fine-tuning methods over 70 diverse decision tasks on Qwen series.

H.2 Fine-tuning Based Methods

To establish a baseline for our investigation into the limitations of global fine-tuning based mitigation, we construct a balanced counterfactual dataset and retrain the model, similar to Zmigrod et al. [58], we initiate this investigation from the specific context of resume screening to evaluate the model's ability to generalize fairness from a singular training domain to unseen decision-making contexts. Specifically, we utilize the Resumes dataset and construct its fairness counterfactuals. We then use this balanced corpus to fine-tune the model. Subsequently, we investigate the fairness performance of this fine-tuned model across 70 distinct decision scenarios on Anthropic Dataset. As illustrated in Table 2 and Figure 8 and A14, the results reveal that even after fairness-aware fine-tuning on the resume dataset, the model still exhibits discrimination in unseen scenarios and shows high volatility across different scenarios.

From a mechanistic perspective, the generalization failure of black-box methods (Global) is attributed to **circuit re-activation**. Since global fine-tuning applies diffuse updates across all parameters to minimize loss on the training set, it fails to effectively inhibit the intrinsic activation of the identified discriminatory components. Consequently, when exposed to unseen decision-making scenarios, these critical components are prone to re-activation, thereby regenerating discriminatory signals.

To empirically validate this hypothesis, we analyze a specific unseen scenario where Global fails to mitigate bias in Figure 8. Figure A13 visualizes the layer-wise evolution of the Discriminatory Intensity ($\mathcal{I}_{l,h}$) for the heads in Llama-8B under both Global and our `pFairFT`. As shown by the red circles, the key heads in the Global model exhibit a high state of expression, particularly in the deeper layers. This confirms that the underlying "Identify-then-Decide" circuit remains intact and active, driving the model to discriminate despite the fine-tuning efforts. In contrast, `pFairFT` (represented by stars) maintains the discriminatory intensity of these key heads at a low level throughout the network. This demonstrates that `pFairFT` effectively prevents the re-activation of bias-related circuits even in novel contexts, thus ensuring fairness generalization.

These findings demonstrate that surface-level interventions are insufficient to guarantee fairness. By treating the model as a black box, these methods fail to address the underlying mechanism of bias, leading to fragility and limited transferability.

I Details on Precise Fairness Fine-Tuning

I.1 Experimental Setup

To assess the effectiveness of our `pFairFT`, we adopt a two-stage evaluation. We first perform the fine-tuning and primary evaluation on the Resume Screening dataset, which serves as the source domain. Subsequently, to assess the method's capability to generalize fairness across unseen domains, we conduct evaluation on the Anthropic benchmark, testing the model's performance across 70 diverse decision-making scenarios without any further parameter updates. We compare `pFairFT` against three baselines: i) Base: uses original, serving as the reference point for intrinsic bias; ii) Global: fine-tunes the whole parameters with counterfactually fair dataset; and iii) `pFairFT`-KL: updates only the key heads with a direct DP constraint, which minimizes KL divergence between the model's output probability distributions conditioned on different sensitive groups. We randomly sample $1,000$ instances from the Resume dataset to construct the training set and 100 samples for test set. To ensure computational efficiency and a fair comparison, all methods are implemented using LoRA. All models are trained for 3 epochs with a fixed learning rate and batch size. The specific hyperparameters used during the training process are summarized in Table A4.

Table A4: Hyperparameters for LoRA fine-tuning on RTX 3090.

Hyperparameter	Value
LoRA Rank (r)	8
LoRA Alpha (α)	16
LoRA Dropout	0.1
Number of Epochs	3
Batch Size (per device)	2
Gradient Accumulation Steps	4
Effective Batch Size	8
Learning Rate	2e-5
Max Sequence Length	512
Warmup Steps	500
Weight Decay	0.01
Optimizer	AdamW
Precision	FP16
Training Samples	1,000

I.2 More results on Other LLMs

We further show the fairness performance of compared fine-tuning methods (Qwen series) over 70 diverse decision tasks in Figure A14.

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